

Forestry and the Environment

Science

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Forestry and the Environment (A06741)

Short Title: Forestry and the Environment
Faculty Science and Computing
Department Science
Cluster Forestry
Author GFEWER
Credits 5

Level: Introductory

Description of Module / Aims

This module recognises forests as more than simply timber production systems. Land use in Ireland past and present and the impacts of forestry on the landscape, and on water, soil, biodiversity and archaeological heritage are discussed. Climate change, forest habitats, biodiversity and the recreation and amenity potential of forests are also covered. An important component of the module is carrying out assignments on forest recreation, archaeological heritage, forest landscape design planning and forest ecosystem services.

Programmes

ENVI-0004 BSc in Forestry (WD_SFORS_D)

2 / 3 / M

Indicative Content

- Forest interactions with climate, soil, water and biodiversity
- Forestry as a land use and its impact on landscape and archaeological heritage
- Forest management for recreation and urban forestry; forest ecosystem services
- Forestry and sustainable rural development, farm forestry
- Irish forests and its stakeholders

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Employ forest landscape design techniques which conform to statutory environmental safeguards regarding forest recreation, archaeological heritage, water and ecology.
2. Describe the complex relationship between forests and their external/internal environments.
3. Determine the ecosystem services provided by forests.
4. Recognise and identify archaeological sites and monuments and the impact such sites have on Forest management.

Learning and Teaching Methods

- Lectures: will be used to describe the knowledge areas covered by the module.
- Field practical: will provide students with the opportunity to gather data for project work.
- Continuous assessment: individual or group assignments in landscape design planning, recreation management, forestry & archaeological heritage, and forest ecology/ecosystem services.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	40%	1
Assignment	30%	2,3
Assignment	30%	4

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self-study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Field Trips	24	
Independent Learning	87	

Essential Material

- Forest Service, F.S. *_Code of best practice_*. Dublin, Ireland: Dept. of Agriculture, Food & Fisheries, 2000.
- Forest Service, F.S. *_Forestry procedures/schemes manual_*. Dublin, Ireland: Dept. of Agriculture, Food & Fisheries, 2004.
- Forest Service, F.S. *_Guidelines on archaeology, water, landscape, harvesting, fertilisation, _protection, biodiversity_*. Dublin, Ireland: Dept. of Agriculture, Food & Fisheries, Various years.

Supplementary Material

- Bell, S. _Designing sustainable forest landscapes_. London, United Kingdom: Taylor & Francis, 2008.
- Forestry Commission, F.C. _Forests and historic environment: UK Forestry Standard Guidelines_. UK: Forestry Commission, 2011.
- Fossit, J. _A guide to habitats in Ireland_. Reprint with 'Notes to Reader' and errata. Kilkenny, Ireland: The Heritage Council, 2007.
- Giller, P., M. Johnson and J. O'Halloran. _Managing the impacts of forest_ _clearfelling__on stream environments_. Dublin, Ireland: COFORD, 2002.
- Johnson, G. _Archaeology and forestry in Ireland_. Dublin, Ireland: The Heritage Council, 1998.
- Thosten, B.J., R. Mavsar, L. Tyrvaenen, I. Prokofieva and E. Stenger. _The provision of ecosystem services: Quantifying and Valuing Non-marketed Ecosystem Services_. Joensuu, Finland: European Forest Institute, 2014.